Phys. 2110 HW # 7 Solution

7. 41, 42. 46. 63. 70. 74. 81. 87

7.41 (a) $s = 280\,ft = 85.3\,m$

The work energy expression gives $\frac{1}{2}mv_1^2 - \mu_k mgs = 0$

$v_1 = \sqrt{2\mu_k gs} = 22.4\,m/s = 50\,mph$; the driver was speeding

(b) 15 mph over speed limit, so \$150 ticket.

7.42 (a) Equating the potential energy stored in the spring to the block's kinetic energy.

$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$. $v = \sqrt{\frac{k}{m}}\,x = \sqrt{\frac{400\,N/m}{2.00\,kg}} \times 0.220\,m = 3.11\,m/s$

(b) Using energy methods directly, the initial potential energy of the spring equals the final gravitational potential energy. $\frac{1}{2}kx^2 = mgL\sin\theta$

$$L = \frac{\frac{1}{2}kx^2}{mg\sin\theta} = \frac{\frac{1}{2}(400\,N/m)(0.220\,m)^2}{(2.00\,kg)(9.80\,m/s^2)\sin 37.0^\circ} = 0.821\,m$$

7.46 (a) Eq.(7.14) applied to points A and B gives $U_A - U_B = \frac{1}{2}mv_B^2$

At B, $a = \frac{v_B^2}{R}$, downward. The minimum speed is when $n \to 0$ and $mg = m\frac{v_B^2}{R}$

The minimum speed required is $v_B = \sqrt{gR}$

Thus $mg(h-2R) > \frac{1}{2}mgR$, $h > \frac{5}{2}R$

(b) Apply Eq (7.14) to points A and C. $U_A - U_C = 2.50\,Rmg = K_C = \frac{1}{2}mv_C^2$, so

$v_C = \sqrt{5.00gR} = \sqrt{(5.00)(9.80\,m/s^2)(20.0\,m)} = 31.3\,m/s$

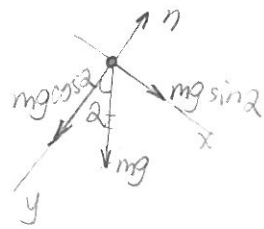
The radial acceleration is $a_{rad} = \frac{v_C^2}{R} = 49.0\,m/s^2$. The tangential direction is down, the normal force at point C is horizontal, there is no friction, so the only downward force is gravity, and $a_{tan} = g = 9.80\,m/s^2$.

7.63

$\Sigma F_y = ma_y$

$mg\cos\alpha - n = m\frac{v_2^2}{R}$

But $n = 0$ so $mg\cos\alpha = m\frac{v_2^2}{R}$ $\quad v_2^2 = Rg\cos\alpha$



Now use conservation of energy to get another equation

$K_1 + U_1 + W_{other} = K_2 + U_2$

The only force that does work on the skier is gravity, so $W_{other} = 0$

$K_1 = 0$, $K_2 = \frac{1}{2}mv_2^2$, $U_1 = mgR$, $U_2 = mgR\cos\alpha$

Then $mgR = \frac{1}{2}mv_2^2 + mgR\cos\alpha$

$v_2^2 = 2gR(1-\cos\alpha)$, combine this with the $\Sigma F_y = ma_y$ equation

$Rg\cos\alpha = 2gR(1-\cos\alpha)$

$\cos\alpha = 2(1-\cos\alpha)$ $\quad \cos\alpha = \frac{2}{3}$ so $\alpha = 48.2^\circ$

7.70 (a) With no friction, $W_{other} = 0$, $K_A = 0$ and $U_A = K_B + U_B$. The maximum speed is when $U_B = 0$ and this is at $x_{1B} = x_{2B} = 0$, when both springs are at their natural length

$$\frac{1}{2}k_1 x_{1A}^2 + \frac{1}{2}k_2 x_{2A}^2 = \frac{1}{2}mv_B^2$$

$x_{1A}^2 = x_{2A}^2 = (0.150\,m)^2$, so $v_B = \sqrt{\frac{k_1+k_2}{m}}(0.150\,m) = \sqrt{\frac{2500\,N/m + 2000\,N/m}{3.00\,kg}}(0.150\,m) = 5.81\,m/s$

(b) At maximum compression of spring 1, spring 2 has its maximum extension and $v_B = 0$. Therefore, at this point $U_A = U_B$. The distance spring 1 is compressed equals the distance spring 2 is stretched: $x_{1A} = -x_{2A}$ and $x_{1B} = -x_{2B}$

Then $U_A = U_B$, $\frac{1}{2}(k_1+k_2)x_{1A}^2 = \frac{1}{2}(k_1+k_2)x_{1B}^2$ and $x_{1B} = -x_{1A} = -0.150\,m$

The maximum compression of spring 1 is 15.0 cm.

7.74 (a) $K_1 = 0$, $U_2 = 0$, $W_{other} = -f_k L = -\mu_k L mg\cos\theta$, $U_1 = mgL\sin\theta$, $K_2 = \frac{1}{2}mv^2$, where v is the speed before the block hits the spring.

Eq. (7.14) applied to points 1 and 2, $U_1 + W_{other} = K_2$. Solving for v

$$v = \sqrt{2gL(\sin\theta - \mu_k\cos\theta)} = \sqrt{2(9.80\,m/s^2)(4.00\,m)(\sin 53.1^\circ - (0.20)\cos 53.1^\circ)} = 7.30\,m/s$$

(b) Apply Eq. (7.14) to points 1 and 3. $K_1 = K_3 = 0$, $U_1 = mg(L+d)\sin\theta$, $U_2 = \frac{1}{2}kd^2$, $W_{other} = -f_k(L+d)$. Eq. (7.14) gives $mg(L+d)\sin\theta - \mu_k mg\cos\theta(L+d) = \frac{1}{2}kd^2$

$$d^2\frac{k}{2mg(\sin\theta - \mu_k\cos\theta)} - d - L = 0$$

Using the quadratic formula gives $d = 1.06\,m$.

(c) If the block ends up a distance y below the starting point, then the block has moved a distance $L+d$ down the incline and $L+d-y$ up the incline. So the work done by friction is $-\mu_k(2L+2d-y)mg\cos\theta$. This must be equal to the change in gravitational potential energy, which is $-mgy\sin\theta$.

$$-\mu_k(2L+2d-y)mg\cos\theta = -mgy\sin\theta$$

$y = (L+d)\frac{2\mu_k}{\tan\theta + \mu_k}$. Using the value of d found in part (b) and the given values for μ_k and θ gives $y = 1.32\,m$.

7.81 (a) $U(r) = -\frac{Gm_1m_2}{r}$, $F_x = -\frac{\partial U}{\partial x} = +Gm_1m_2\frac{\partial(\frac{1}{r})}{\partial x} = -\frac{Gm_1m_2x}{(x^2+y^2+z^2)^{3/2}}$

Similarly, $F_y = -\frac{Gm_1m_2y}{(x^2+y^2+z^2)^{3/2}}$ and $F_z = -\frac{Gm_1m_2z}{(x^2+y^2+z^2)^{3/2}}$

(b) $(x^2+y^2+z^2)^{\frac{3}{2}} = r^3$ so $F_x = -\frac{Gm_1m_2x}{r^3}$, $F_y = -\frac{Gm_1m_2y}{r^3}$, $F_z = -\frac{Gm_1m_2z}{r^3}$

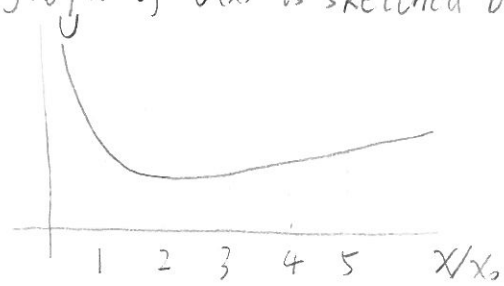
$$F = \sqrt{F_x^2 + F_y^2 + F_z^2} = \frac{Gm_1m_2}{r^3}\sqrt{x^2+y^2+z^2} = \frac{Gm_1m_2}{r^2}$$

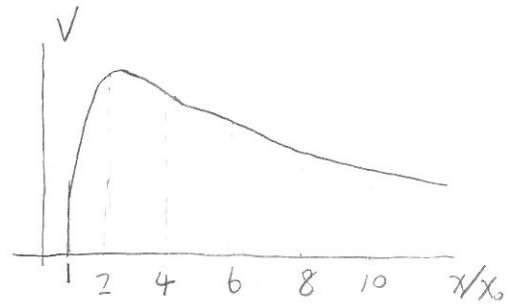
(c) F_x, F_y and F_z are negative. $F_x = \alpha x$, $F_y = \alpha y$ and $F_z = \alpha z$ where α is a constant, so $\vec{F}$ and the vector $\vec{r}$ from m_1 to m_2 are in the same direction. Therefore, $\vec{F}$ is directed toward m_1 at the origin and $\vec{F}$ is attractive.

7.87 (a) $\beta = 2/x_0$ $\quad U(x) = \frac{\alpha}{x^2} - \frac{\beta}{x} = \frac{\alpha}{x_0^2}\frac{x_0^2}{x^2} - \frac{\alpha}{x_0 x} = \frac{\alpha}{x_0^2}\left[\left(\frac{x_0}{x}\right)^2 - \left(\frac{x_0}{x}\right)\right]$

$U(x_0) = \left(\frac{\alpha}{x_0^2}\right)(1-1) = 0$. $U(x)$ is positive for $x < x_0$ and negative for $x > x_0$.

The graph of $U(x)$ is sketched below





(b) $V(x) = \sqrt{-\frac{2}{m}U} = \sqrt{\left(\frac{2\alpha}{m x_0^2}\right)\left(\left(\frac{x_0}{x}\right) - \left(\frac{x_0}{x}\right)^2\right)}$.

The proton moves in the positive x-direction, speeding up until it reaches a maximum speed and the slow down, although it never stops. The minus sign in the square root in the expression for $V(x)$ indicates that the particle will be found only in the region where $U<0$, that is $x > x_0$. The graph of $V(x)$ is sketched above.

(c) The maximum speed corresponds to the maximum kinetic energy, and hence the minimum potential energy.

This minimum occurs when $\frac{dU}{dx} = 0$. $\quad \frac{dU}{dx} = \frac{\alpha}{x_0^2}\left[-2\left(\frac{x_0}{x}\right)^3 + \left(\frac{x_0}{x}\right)^2\right] = 0$

which has the solution $x = 2x_0$. $\quad U(2x_0) = -\frac{\alpha}{4x_0^2}$, so $V = \sqrt{\frac{\alpha}{2m x_0^2}}$

(d) The maximum speed occurs at $\frac{dU}{dx} = 0$, and from $F = -\frac{dU}{dx}$, $F = 0$ at this point.

(e) $x_1 = 3x_0$, and $U(3x_0) = -\frac{2\alpha}{9x_0^2}$

$V(x) = \sqrt{\frac{2}{m}(U(x_1) - U(x))} = \sqrt{\frac{2\alpha}{m x_0^2}\left(\left(\frac{x_0}{x}\right) - \left(\frac{x_0}{x}\right)^2 - \frac{2}{9}\right)}$

The particle is confined to the region where $U(x) < U(x_1)$. The maximum speed still occurs at $x = 2x_0$ but now the particle will oscillate between x_1 and some minimum value.

(f) Note that $U(x) - U(x_1)$ can be written

$$\frac{\alpha}{x_0^2}\left[\left(\frac{x_0}{x}\right)^2 - \left(\frac{x_0}{x}\right) + \frac{2}{9}\right] = \frac{\alpha}{x_0^2}\left[\left(\frac{x_0}{x}\right) - \frac{1}{3}\right]\left[\frac{x_0}{x} - \frac{2}{3}\right]$$

which is zero at $x = 3x_0 = x_1$ and $x = \frac{3}{2}x_0$. Thus, when the particle is released from x_0, it goes on to infinity and doesn't reach any maximum distance. When released from x_1, it oscillates between $\frac{3}{2}x_0$ and $3x_0$.